

MongoDB Aggregation — Terror Track

30 Practice Problems — sample_restaurants

No answers. Use PyMongo and the aggregation operators covered so far.

- Q1.** Count the number of restaurants in each borough.
- Q2.** Unwind the grades array and display the resulting documents.
- Q3.** Find the average grade score for each borough.
- Q4.** Find the number of restaurants for each cuisine in Brooklyn.
- Q5.** Find the highest grade score recorded in each borough.
- Q6.** Find the lowest grade score recorded in each borough.
- Q7.** Find the average grade score for each cuisine, considering only restaurants in Manhattan.
- Q8.** Find the number of restaurants for each cuisine in Queens, sorted by count from highest to lowest.
- Q9.** Find the average score for each cuisine in Queens, considering only grades with score greater than 10, and sort by average score descending.
- Q10.** Count the number of unique restaurants in each borough that have received at least one grade of A.
- Q11.** Count the number of unique restaurants in each cuisine that have at least one grade with score greater than 4.
- Q12.** For Brooklyn, find the maximum grade score for each cuisine, considering only grades with score less than 8. Sort by maximum score descending.
- Q13.** For Queens, find the minimum grade score for each cuisine, considering only grades with score greater than 3. Sort by minimum score ascending.
- Q14.** Considering only grades with score greater than 5 and grade A, find the average score for each borough and sort by average score descending.
- Q15.** Count the number of unique restaurants in each borough that have received a grade with score exactly 10.
- Q16.** Find the average score for each cuisine, but count each restaurant only once for the calculation. First calculate each restaurant's average, then average those restaurant averages by cuisine.
- Q17.** Find the number of unique restaurants in each borough that have at least one grade with score greater than 15.
- Q18.** For Manhattan, find the number of unique restaurants for each cuisine that have at least one grade A with score greater than 8.
- Q19.** Find the highest grade score for each cuisine in Brooklyn, considering only restaurants that have at least one grade A.

- Q20.** Find the lowest grade score for each borough among restaurants whose cuisine is American.
- Q21.** Find the average score for each borough among restaurants whose cuisine is Italian, considering all grades.
- Q22.** Find the number of unique restaurants for each cuisine that have at least one grade B or C.
- Q23.** Find the average score for each borough for grades A, then return only the three boroughs with the lowest average score.
- Q24.** Find the number of unique restaurants in each borough that have at least one grade with score greater than 12, and sort by the number of restaurants descending.
- Q25.** For Queens, find the average score for each cuisine after excluding grades with score 10 or less.
- Q26.** Find the maximum score for each borough for restaurants whose cuisine is American or Italian.
- Q27.** Find the number of unique restaurants in each cuisine that have at least one grade A with score greater than 7, then sort by count descending.
- Q28.** Find the average grade score for each borough, but first calculate the average score for each individual restaurant and then average those restaurant averages by borough.
- Q29.** Find the three cuisines with the highest average grade score in Manhattan, considering only grades with score greater than 5.
- Q30.** Find the number of unique restaurants in each borough that have both an A grade and at least one grade with score greater than 15.